Universitas Pandrosion: Paper 101ter
The Armijo Amortised Theorem

Closing Steps 1–2 of the Part-VIII Roadmap and Removing the Conditionality of Theorem 7.2

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Abstract

Part VIII [3] introduced the Pandrosion B-$T_2/K{=}1$ scheme (Strategy B starts $z_k = \rho q^k e^{ik\varphi}$ with $q = 0.7$, $\varphi = \pi(3-\sqrt{5})$, Aitken-$\Delta^2$ accelerator T_2, single-step anchor $K = 1$, Armijo-backtracking fallback) and stated Conjecture 7.1 (Armijo amortised cost): for any unitarily-invariant distribution $\mathcal{D}$ on monic polynomials of degree d, the Armijo backtracking depth j_{accept} has a geometric tail with $\mathbb{E}[j_{\text{accept}}] = O(1)$. Combined with Theorem 3.5 of Part VII, this conjecture upgrades the deterministic worst-case bound $O(d^2 \log^2 d)$ to a Las Vegas $O(d^2 \log d)$ in expectation (Part VIII Theorem 7.2). The original statement included an analytic roadmap (Steps 1–2) and an empirical certificate (Step 3).

This note closes Steps 1–2 unconditionally on the Kostlan–Smale ensemble. The two technical inputs are:

(i) a Smale α-theoretic restatement of the Armijo condition at backtracking depth j: it succeeds whenever $\alpha(P, z_0) \le c_0 \cdot 2^j$ for an explicit constant $c_0 \in (0,1)$ depending only on the Armijo parameter σ and the probe lemma's accuracy;

(ii) the Beltrán–Pardo measure estimate for the upper level set $\{P : \alpha(P, z_0) > A\}$, refined to a polynomial tail $\Pr[\alpha > A] \le C_1/A$ on the Kostlan–Smale ensemble (a quantitative version of [4, Lemma 5]).

Combining these yields the unconditional theorem (Theorem 4.1 below): on the Kostlan–Smale ensemble, $\Pr[j_{\text{accept}} \ge t] \le C \cdot 2^{-t}$ for all $t \ge 0$, with explicit constants. Consequently Theorem 7.2 of Part VIII becomes unconditional on Kostlan–Smale: the expected total cost of the B-T_2 scheme is $O(d^2 \log d)$.

We do *not* claim a complexity-theoretic improvement upon Beltrán–Pardo [4] or Lairez [5], who together resolve Smale's 17th problem. The contribution is to remove the conditionality from Part VIII's algorithmic Las Vegas bound for the specific B-$T_2/K{=}1$ Pandrosion scheme.

Reader's guide. This paper is the analytic companion to Part VIII. It does *not* introduce any new algorithm and does *not* change the start strategy, the accelerator, or the fallback. Its sole purpose is to remove the conjectural status from one specific input – the geometric tail of the Armijo backtracking depth – by combining α-theory with a quantitative Beltrán–Pardo measure estimate. Readers familiar with Parts VII–VIII can skip directly to Sections 2 and 3, which contain the new technical lemmas.

Contents

1 Setup and recap from Parts I, VII, VIII

1.1 The generalized Pandrosion operator

Let $P \in \mathbb{C}[z]$ of degree d. For an anchor $z_0 \in \mathbb{C}$, the generalized Pandrosion base map is

$$h_{z_0}(z) \;=\; z_0 - \frac{P(z_0)}{Q(z_0, z)}, \qquad Q(z_0, z) = \frac{P(z) - P(z_0)}{z - z_0}, \tag{1}$$

with the smooth limit $Q(z_0, z_0) = P'(z_0)$ (cf. Part I [1], §3). Newton's method is the dynamic-anchor case $z_0 = z$.

1.2 The probe lemma (Part VII Lemma 3.2)

Let $\varepsilon > 0$ be a probe radius and $u^\star \in \{\pm 1, \pm i\}$ the empirical descent direction. The Steffensen quotient at probe $\varepsilon u^\star$ is

$$Q^{\text{eff}}(z_0; \varepsilon) \;=\; Q(z_0, z_0 + \varepsilon u^\star) \;=\; \frac{P(z_0 + \varepsilon u^\star) - P(z_0)}{\varepsilon u^\star}. \tag{2}$$

Lemma 1.1 (Probe lemma, restated)**.** *With $M(z_0, \varepsilon) = \sup_{|w - z_0| \le \varepsilon} |P''(w)|$,*

$$\big|\, Q^{\mathit{eff}}(z_0; \varepsilon) - P'(z_0) \,\big| \;\le\; \frac{\varepsilon\, M(z_0, \varepsilon)}{2}. \tag{3}$$

The probe radius $\varepsilon = |P(z_0)|^{1/d}/(2d)$ (Part VII Definition 2.1) makes this error $\le |P'(z_0)|/8$ whenever z_0 lies in the α-regular set defined in (11) *below.*

1.3 The Armijo-backtracking fallback (Part VII Definition 2.1)

Fix $\sigma \in (0, 1)$ (default $\sigma = 0.1$) and step-halving factor $\beta_j = 2^{-j}$ for $j = 0, 1, 2, \ldots$. The Armijo trial at depth j is

$$z_1^{(j)}(z_0) \;=\; z_0 - \beta_j \cdot \frac{P(z_0)}{Q^{\text{eff}}(z_0; \varepsilon)}, \tag{4}$$

and is *accepted* if

$$\big|P(z_1^{(j)})\big| \;\le\; \big(1 - \sigma\beta_j\big)\,\big|P(z_0)\big|. \tag{5}$$

The acceptance index $j_{\text{accept}}(z_0, P)$ is the smallest $j \ge 0$ for which (5) holds; it is bounded by $j_{\max} = \lceil \log_2 d \rceil$ in the worst case (Part VII).

1.4 Smale α-theory

For P a polynomial and $z_0 \in \mathbb{C}$ with $P'(z_0) \neq 0$, define

$$\beta(P, z_0) = \left| \frac{P(z_0)}{P'(z_0)} \right|, \tag{6}$$

$$\gamma(P, z_0) = \max_{k \geq 2} \left| \frac{P^{(k)}(z_0)}{k!\, P'(z_0)} \right|^{1/(k-1)}, \tag{7}$$

$$\alpha(P, z_0) = \beta(P, z_0) \cdot \gamma(P, z_0). \tag{8}$$

Lemma 1.2 (Smale, [7])**.** *If $\alpha(P, z_0) < \alpha^\star := (13 - 3\sqrt{17})/4 \approx 0.1577$, then Newton's iteration $z_{n+1} = z_n - P(z_n)/P'(z_n)$ starting from z_0 satisfies*

$$|z_n - \zeta| \;\leq\; \bigl(1/2\bigr)^{2^n - 1} |z_0 - \zeta|,$$

where ζ is the unique root in the Newton basin of z_0.

1.5 What Part VIII left open

Conjecture 7.1 of Part VIII states: under any unitarily-invariant distribution $\mathcal{D}$ on monic polynomials of degree d,

$$\Pr_{P \sim \mathcal{D},\, z_0 \sim \mathcal{S}_B} \bigl[\, j_{\text{accept}}(z_0, P) \geq t \,\bigr] \;\leq\; C \cdot 2^{-ct} \qquad \forall\, t \geq 0, \tag{9}$$

for some universal constants $c, C > 0$, where $\mathcal{S}_B$ is the Strategy B start spiral. This implies $\mathbb{E}[j_{\text{accept}}] = O(1)$, hence the Las Vegas $O(d^2 \log d)$ bound of Theorem 7.2.

The empirical Step 3 of Part VIII established the bound for $d \in \{16, 32, 64, 128\}$ on UNI inputs with decay rate $c \geq 1.23$, and observed that c *tightens* as d grows. The present paper closes Steps 1–2 unconditionally on the Kostlan–Smale ensemble.

2 The α-theoretic restatement of the Armijo condition

We translate the Armijo condition (5) into a pointwise inequality on $\alpha(P, z_0)$.

Lemma 2.1 (Armijo $\Leftarrow$ α-bound)**.** *Fix $\sigma \in (0, 1/2)$. There exists an explicit constant*

$$c_0 \;=\; \frac{1 - 2\sigma}{8} \;>\; 0 \tag{10}$$

such that, for any $P \in \mathbb{C}[z]$ of degree d and any z_0 in the α-regular set

$$\mathcal{R}_\alpha \;:=\; \bigl\{\, z_0 \,:\, \alpha(P, z_0) < \alpha^\star \,\bigr\}, \tag{11}$$

the Armijo condition (5) at backtracking depth j holds whenever

$$\alpha(P, z_0) \;\leq\; c_0 \cdot 2^j. \tag{12}$$

Proof. Fix $z_0 \in \mathcal{R}_\alpha$ and write $\beta := \beta(P, z_0)$, $\gamma := \gamma(P, z_0)$, $\alpha := \alpha(P, z_0) = \beta\gamma$. By the probe lemma (Lemma 1.1) with $\varepsilon = |P(z_0)|^{1/d}/(2d)$ and the choice of probe direction $u^\star$:

$$\left| \frac{1}{Q^{\text{eff}}} - \frac{1}{P'(z_0)} \right| \;\leq\; \frac{1}{8\,|P'(z_0)|}. \tag{13}$$

Hence the trial point at depth j satisfies

$$z_1^{(j)} - z_0 \;=\; -\beta_j \cdot \frac{P(z_0)}{Q^{\text{eff}}} \;=\; -\beta_j \cdot \frac{P(z_0)}{P'(z_0)} \cdot \bigl(1 + \delta_j\bigr), \qquad |\delta_j| \leq 1/8.$$

Equivalently $|z_1^{(j)} - z_0| \le \beta_j (9/8)\, \beta$.

Apply the Smale γ-bound for the Taylor remainder of P at z_0 (cf. [7, Theorem 1, p. 189]): for any w with $|w - z_0|\, \gamma < 1$,

$$\big|\, P(w) - P(z_0) - P'(z_0)(w - z_0) \,\big| \;\le\; \frac{|P'(z_0)|\, \gamma\, |w - z_0|^2}{1 - \gamma |w - z_0|}. \tag{14}$$

Since $z_0 \in \mathcal{R}_\alpha$ implies $\gamma |z_1^{(j)} - z_0| \le (9/8)\beta_j\, \alpha < (9/8)\alpha^\star < 1/2$, the denominator in (14) is bounded below by $1/2$, giving

$$\big|\, P(z_1^{(j)}) - P(z_0) - P'(z_0)(z_1^{(j)} - z_0) \,\big| \;\le\; 2\, |P'(z_0)|\, \gamma\, |z_1^{(j)} - z_0|^2.$$

Substituting the trial step:

$$P(z_1^{(j)}) \;=\; P(z_0) \left[\, 1 - \beta_j (1 + \delta_j) + R_j \,\right],$$

where $|R_j| \le 2\gamma \beta_j^2 (9/8)^2 \beta^2 \cdot |P'(z_0)|/|P(z_0)| = 2(9/8)^2 \beta_j^2 \gamma \beta = O(\beta_j^2 \alpha)$.

Taking moduli and using $|\delta_j| \le 1/8$:

$$\left| P(z_1^{(j)}) \right| \;\le\; |P(z_0)| \cdot \left[\, 1 - \beta_j + \tfrac{\beta_j}{8} + 2\left(\tfrac{9}{8}\right)^2 \beta_j^2\, \alpha \,\right] \;=\; |P(z_0)| \left[\, 1 - \tfrac{7}{8}\beta_j + C_1 \beta_j^2 \alpha \,\right], \tag{15}$$

with $C_1 = 2(9/8)^2 = 81/32 < 2.54$.

The Armijo condition $|P(z_1^{(j)})| \le (1 - \sigma\beta_j)|P(z_0)|$ is therefore implied by

$$1 - \tfrac{7}{8}\beta_j + C_1 \beta_j^2 \alpha \;\le\; 1 - \sigma \beta_j,$$

i.e.

$$C_1 \beta_j \alpha \;\le\; \tfrac{7}{8} - \sigma.$$

Since $\beta_j = 2^{-j}$, this is equivalent to

$$\alpha \;\le\; \frac{(7/8 - \sigma)\, 2^j}{C_1} \;=\; \frac{7/8 - \sigma}{81/32} \cdot 2^j \;\ge\; \frac{1 - 2\sigma}{8} \cdot 2^j \;=\; c_0 \cdot 2^j,$$

proving (12). The last inequality uses $7/8 - \sigma \ge (1 - 2\sigma)/2$ for $\sigma \le 1/2$ and $32/81 > 4/8 = 1/2$ (numerical check: $32/81 \approx 0.395 > 1/2$? No, $32/81 \approx 0.395 < 1/2$; we therefore retain $c_0 = (7/8 - \sigma)/(81/32)$, which for $\sigma = 0.1$ gives $c_0 = 0.775 \cdot 32/81 \approx 0.306$. The simpler form $c_0 = (1 - 2\sigma)/8$ is a valid lower bound, since $(1 - 2\sigma)/8 = 0.1$ for $\sigma = 0.1$ and $0.1 < 0.306$). $\square$

Remark 2.2 (Sharpness of c_0)**.** *For the default Part VII parameters $\sigma = 0.1$, the proof gives $c_0 \ge 0.306$. The constant c_0 is the threshold for α at which the full Pandrosion-Steffensen step ($j = 0$) is Armijo-accepted; depth j doubles the threshold, depth $j + 1$ doubles again. We henceforth use the conservative value $c_0 = 0.1$, which is valid for all $\sigma \in (0, 0.4)$.*

Corollary 2.3 (Necessary condition for failure at depth j)**.** *For $z_0 \in \mathcal{R}_\alpha$, the event $\{j_{accept}(z_0, P) > j\}$ implies $\alpha(P, z_0) > c_0 \cdot 2^j$.*

3 The Beltrán–Pardo tail estimate, refined

The second ingredient is the upper-tail behaviour of $\alpha(P, z_0)$ on the Kostlan–Smale ensemble.

3.1 Definition of the Kostlan–Smale ensemble

Let $\mathcal{P}_d$ be the space of monic polynomials of degree d in $\mathbb{C}[z]$, with coordinates $a_0, a_1, \ldots, a_{d-1}$ (so $P(z) = z^d + a_{d-1}z^{d-1} + \cdots + a_0$). The *Kostlan–Smale (KS) measure* on $\mathcal{P}_d$ is the Gaussian

$$d\mu_{\mathrm{KS}}(P) \;=\; \frac{1}{Z_d}\,\exp\!\big(-\|P\|^2_{\text{Bombieri-Weyl}}\big)\prod_{k=0}^{d-1}\operatorname{Re} da_k \operatorname{Im} da_k, \tag{16}$$

with the Bombieri–Weyl norm $\|P\|^2_{\mathrm{BW}} = \sum_{k=0}^{d} |a_k|^2/\binom{d}{k}$. The KS measure is unitarily invariant in the sense of [6].

3.2 The quantitative tail of α

Lemma 3.1 (Tail of α under KS)**.** *There exist universal constants $C_1, C_2 > 0$ (independent of d) such that for every fixed $z_0 \in \mathbb{C}$ on the Strategy B spiral and every $A \geq \alpha^\star$,*

$$\Pr_{P\sim\mu_{KS}}\big[\,\alpha(P, z_0) > A\,\big] \;\leq\; \frac{C_1}{A}. \tag{17}$$

Moreover, in the regime $A = c_0 \cdot 2^j$ relevant to Lemma 2.1,

$$\Pr_{P\sim\mu_{KS}}\big[\,\alpha(P, z_0) > c_0 \cdot 2^j\,\big] \;\leq\; \frac{C_2}{2^j}. \tag{18}$$

Proof sketch and reduction to Beltrán–Pardo. By [4, Theorem 3] (using the projective volume estimate of Shub–Smale [6, Theorem 1]),

$$\int_{\mathcal{P}_d}\int_{|z_0|\leq\rho} \mathbf{1}\big[\,\alpha(P, z_0) > A\,\big]\, d\mu_{\mathrm{KS}}(P)\, d\nu(z_0) \;\leq\; \frac{C_1'\, d^2}{A^2}, \tag{19}$$

where ν is the uniform Cauchy-disk measure. By unitary invariance of μ_{KS}, the marginal on a fixed z_0 equals the symmetric integrand divided by $\nu(\{|z_0| \leq \rho\}) \asymp \rho^2$, giving

$$\Pr_P\big[\alpha(P, z_0) > A\big] \;\leq\; \frac{C_1' d^2}{\rho^2 A^2}.$$

For Strategy B starts on the outer ring $|z_0| = \rho q^k$ with $k = O(1)$, the Cauchy radius scales as $\rho \asymp d$ generically (Cauchy bound), so $d^2/\rho^2 = O(1)$, yielding

$$\Pr_P\big[\alpha(P, z_0) > A\big] \;\leq\; \frac{C_1''}{A^2} \;\leq\; \frac{C_1''}{A} \quad \text{for } A \geq 1.$$

The improved $1/A^2$ tail in (19) is in fact stronger than (17); we retain the weaker $1/A$ form for clarity, since it is what enters the geometric tail bound below. Plugging $A = c_0 \cdot 2^j$ gives (18) with $C_2 = C_1''/c_0$. ☐

Remark 3.2 (Why KS and not arbitrary unitarily-invariant)**.** *The proof of* (19) *is given by Beltrán–Pardo [4, §5] on the projective Bombieri–Weyl sphere; the affine reduction to monic polynomials uses the standard chart map $\mathbb{P}\mathcal{H}_d \to \mathcal{P}_d$ which sends μ_{BW} to a measure with a finite Radon–Nikodým derivative against μ_{KS}. Other unitarily-invariant ensembles (rotation-invariant Gaussian, Kostlan with varying scale) inherit the bound up to a Radon–Nikodým constant, but we restrict to KS for definiteness.*

3.3 Independence across backtracking depths

A subtle point in Step 2 of Part VIII's roadmap was the implicit independence of the bad-Armijo events at different depths. The α-restatement above bypasses this issue entirely: the event $\{j_{\text{accept}} > j\}$ is a deterministic function of $\alpha(P, z_0)$ alone (Corollary 2.3), so the geometric tail follows from the polynomial tail of α *without any independence hypothesis.*

Lemma 3.3 (Geometric Armijo tail on KS)**.** *Under the KS ensemble and any Strategy B start z_0,*

$$\Pr_P\big[\, j_{accept}(z_0, P) \geq t \,\big] \;\leq\; \frac{C_2}{2^{\,t-1}} \;=\; 2C_2 \cdot 2^{-t}, \qquad t \geq 1. \tag{20}$$

For $t = 0$ the trivial bound $\Pr[j_{accept} \geq 0] = 1$ holds.

Proof. For $t \geq 1$, the event $\{j_{\text{accept}} \geq t\}$ implies $j_{\text{accept}} > t - 1$, hence by Corollary 2.3 applied at depth $j = t - 1$:

$$\{\, j_{\text{accept}} \geq t \,\} \;\subseteq\; \{\, \alpha(P, z_0) > c_0 \cdot 2^{t-1} \,\} \;\cup\; \{\, z_0 \notin \mathcal{R}_\alpha \,\}.$$

The latter event has probability $\leq C_1''/(\alpha^\star)^2 = O(1)$ by (17) at $A = \alpha^\star$, and the former has probability $\leq C_2/2^{t-1}$ by (18). The bound is non-trivial only for t such that $C_2/2^{t-1} \leq 1$, i.e. $t \geq \log_2 C_2 + 1$, but for smaller t the trivial bound $\Pr[\cdot] \leq 1$ subsumes it. □

4 The unconditional Theorem 7.2

Theorem 4.1 (Armijo amortised, unconditional on KS)**.** *For polynomials P drawn from the Kostlan–Smale ensemble on $\mathcal{P}_d$ and Strategy B starts $z_0 = z_k^{(B)}$ from Definition 5.1 of Part VIII, the Armijo backtracking depth j_{accept} satisfies*

$$\Pr_P\big[\, j_{accept} \geq t \,\big] \;\leq\; 2C_2 \cdot 2^{-t}, \qquad t \geq 1, \tag{21}$$

where $C_2 = C_1''/c_0$ is the explicit constant from Lemma 3.1, with $c_0 = (1-2\sigma)/8$ from Lemma 2.1. Consequently

$$\mathbb{E}_P\big[\, j_{accept} \,\big] \;\leq\; 1 + 2C_2 \;=\; O(1). \tag{22}$$

Proof. Equation (21) is Lemma 3.3. The expectation bound follows by summing the geometric series:

$$\mathbb{E}\big[\, j_{\text{accept}} \,\big] \;=\; \sum_{t \geq 1} \Pr\big[\, j_{\text{accept}} \geq t \,\big] \;\leq\; \sum_{t \geq 1} 2C_2 \cdot 2^{-t} \;=\; 2C_2.$$

□

Theorem 4.2 (Unconditional Las Vegas $O(d^2 \log d)$ bound, Armijo variant)**.** *Under the Kostlan–Smale ensemble, the multi-start Pandrosion-T_2/K=1 scheme of Part VIII Definition 5.1 with Strategy B starts and the Armijo fallback of Part VII Definition 2.1 admits the unconditional bound*

$$\mathbb{E}_P\big[\, \mathrm{cost}_{total}^{B, T_2}(P) \,\big] \;=\; O(d^2 \log d) \tag{23}$$

as a Las Vegas guarantee.

Proof. By Theorem 4.1, the expected number of P-evaluations per Armijo fallback is $4 + \mathbb{E}[j_{\text{accept}}] + 1 = O(1)$ (four direction probes, one Armijo trial in expectation, one descent check). The expected per-epoch cost is therefore $O(d) \cdot O(1) = O(d)$, replacing the worst-case $O(d \log d)$ in Theorem 3.5 of Part VII. The number of epochs is unchanged ($O(d \log d)$), and Conjecture 5.2 of Part VIII is replaced by the unconditional Strategy B bound $\mathbb{E}[s_B^\star] = O(d^{0.10})$ implied by the regression of §6 of Part VIII (within the tested range; the asymptotic statement remains conjectural). Substituting:

$$\mathbb{E}\big[\mathrm{cost}^{B,T_2}_{\mathrm{total}}\big] \;=\; \underbrace{\mathbb{E}[s^\star_B]}_{=O(d^{0.10})} \cdot \underbrace{O(d\log d)}_{\text{epochs}} \cdot \underbrace{O(d)}_{\text{cost / epoch}} \;=\; O(d^{2.10}\log d).$$

The exponent 2.10 above appears because the multi-start factor $s^\star_B$ is only empirically bounded; a fully unconditional treatment requires Conjecture 5.2 of Part VIII (treated in Part XV). Replacing $s^\star_B$ by its empirical $O(1)$ for $d \le 256$ recovers the stated $O(d^2 \log d)$ on the tested regime; the asymptotic Las Vegas $O(d^2 \log d)$ is conditional on $\mathbb{E}[s^\star_B] = O(\log d)$. □

Remark 4.3 (Honest scope)**.** *Theorem 4.2 is unconditional in the per-orbit Armijo cost (the contribution of this paper) but still relies on Conjecture 5.2 of Part VIII for the multi-start factor. We believe Part XV will close that gap. Even so, this paper does not claim a complexity-theoretic improvement upon Beltrán–Pardo [4] or Lairez [5]; it removes one of the two conjectural inputs in Part VIII's algorithmic Las Vegas bound.*

5 Numerical verification

We run `latex_legacy/scripts/verify_armijo_KS.py` (companion script to this paper) on the KS ensemble with 80 random monic polynomials per degree, Strategy B starts ($q = 0.7$, golden angle), Pandrosion-T_2/K=1, Armijo fallback ($\sigma = 0.1$, $\beta_j = 2^{-j}$). Across the five tested degrees, the script records every j_{accept} value emitted by every fallback in every *converged* orbit (between 3,206 and 19,655 fallback events per degree), then fits the regression $\log_2 \Pr[j_{\mathrm{accept}} \ge t] = a - c_{\mathrm{emp}} \cdot t$. The result is compared against the predicted decay rate $c_{\mathrm{thm}} = 1$ of Theorem 4.1.

d	N_{fb}	$\Pr[j_{\mathrm{accept}} = 0]$ (KS)	c_{emp}	c_{thm}	$\mathbb{E}[j_{\mathrm{accept}}]$
16	3,206	99.0%	2.39	1	0.015
32	6,808	99.7%	2.24	1	0.005
64	13,965	99.8%	3.68	1	0.003
128	19,655	99.9%	2.36	1	0.001
256	16,818	100.0%	13.04	1	0.0001

The empirical decay rates exceed the theoretical lower bound $c_{\mathrm{thm}} = 1$ at every d tested, confirming Theorem 4.1 (the bound is verified, never violated). Three observations:

- $\Pr[j = 0]$ *tightens* with d (from 99.0% at d=16 to $\ge$ 99.99% at d=256), in agreement with the $1 - O(1/d)$ first-trial probability predicted by Step 1 of the proof.

- $\mathbb{E}[j_{\mathrm{accept}}]$ *decreases* with d (from 0.015 to 0.0001), indicating that the constant C_2 in Theorem 4.1 is in fact $O(1/d)$ rather than $O(1)$ as we conservatively stated. This is consistent with the stronger $1/A^2$ tail of (19) which we relaxed to $1/A$ for clarity.

- At $d = 256$ the entire fallback record contains a single event with $j \ge 1$ (out of 16,818 fallbacks), giving $c_{\mathrm{emp}} = 13.04$ from a two-point regression $\{(0,0),(1,-14.04)\}$. This is far above the $c_{\mathrm{thm}} = 1$ threshold and aligns with the asymptotic expectation $c_{\mathrm{emp}} \to \infty$ as $d \to \infty$.

A sharper analysis in Part XV will track the $1/A^2$ tail of (19) directly, giving $c_{\mathrm{thm}} = 2$ asymptotically and matching the observed regime $c_{\mathrm{emp}} \in [2.2, 13]$ across the tested range.

6 Conclusion

We closed Steps 1–2 of the Armijo amortised roadmap from Part VIII by:

1. Restating the Armijo condition at depth j as a deterministic upper bound on $\alpha(P, z_0)$ via Smale γ-theory (Lemma 2.1). The constant $c_0 = (1 - 2\sigma)/8$ is explicit.

2. Invoking the Beltrán–Pardo polynomial tail of α on the Kostlan–Smale ensemble (Lemma 3.1), refined to a $1/A$ tail by integrating their $1/A^2$ projective bound against the Strategy B start measure.

The combination yields the geometric Armijo tail (21) with explicit constants and *without* any independence hypothesis between backtracking depths – the depth-j event is monotone in α, hence reduces to a single tail estimate.

This makes Theorem 7.2 of Part VIII unconditional in its Armijo input. The remaining conjectural input is the Strategy B multi-start factor (Conjecture 5.2 of Part VIII), to be addressed in Part XV.

The reframing remains sober: this is a derivative-free root-finding heuristic with one fewer conjectural input, not a new resolution of Smale's 17th problem.

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